

Question 1 – Linux Environment and File Management

```
Last login: Fri Aug 14 18:57:21 on tty8010
abhijeetsingh@Abhijeets-MacBook-Air ~ % find ~ -type d -name "MediVerify-AI" 2>/dev/null
/Users/abhijeetsingh/MediVerify-AI/MediVerify-AI
abhijeetsingh@Abhijeets-MacBook-Air ~ % % mkdir Cloud-Assignment
abhijeetsingh@Abhijeets-MacBook-Air ~ % cd Cloud-Assignment
abhijeetsingh@Abhijeets-MacBook-Air Cloud-Assignment % pwd
/Users/abhijeetsingh/Cloud-Assignment
abhijeetsingh@Abhijeets-MacBook-Air Cloud-Assignment % mkdir Abhijeet-GLA-CourseRegistration
abhijeetsingh@Abhijeets-MacBook-Air Cloud-Assignment % cd Abhijeet-GLA-CourseRegistration
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % pwd
/Users/abhijeetsingh/Cloud-Assignment/Abhijeet-GLA-CourseRegistration
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % mkdir src docs backup
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % ls
backup  docs      src
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % touch course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % ls
backup      course_info.txt  docs
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % nano course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % cat course_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
subject name:cloud computing

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % tree
zsh: command not found: tree
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % brew install tree
=> Auto-updating Homebrew...
Adjust how often this is run with '$HOMEBREW_AUTO_UPDATE_SECS' or disable with
'$HOMEBREW_NO_AUTO_UPDATE=1'. Hide these hints with '$HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
=> Auto-updated Homebrew!
Updated 2 taps (homebrew/core and homebrew/cask).
=> New Formulae
  auto-updater: Multi-platform desktop utility for coding agents
  mongrel: Database workbench with terminals, containers, Kubernetes, and API client
  motrix@beta: Open-source download manager
  muse-code: Interactive terminal coding agent
  nativ: Run AI models locally
  orbit: Multiple Google accounts in isolated sessions in one window
  owlcr: On-device OCR for screenshots, images, and PDFs
  petdex: Desktop pet that reflects coding agent activity
  phosphene: Custom video wallpapers for the desktop and lock screen
  pixcake: AI photo editing software for commercial photography
  quarkcloudrive: Cloud storage and file management platform
  redot: Multi-platform 2D and 3D game engine
  riverscript: AI platform for recording and transcribing system audio from any app
  sentry-cli: Command-line utility to interact with Sentry
  sina-finance: Stock market data and financial news platform
  sjmcl: Minecraft launcher built with the community
  skills-manager: Manage, sync, and organise AI agent skills across coding tools
  space-rabbit: Removes animations when switching between Spaces
  spacejump: Menu bar utility to name and switch desktop Spaces
  subtitle-edit: Subtitle editor
  switchy: Switch Magic Keyboard, Trackpad and Mouse between Macs
  tabularis: Lightweight database management tool
  tauritavern: Sillyfavern-compatible native client
  top-viewer: Packet capture and inspection tool
  textream: Teleprompter that highlights scripts in real time as you speak
  thoughtdag: Visual workspace for editable LLM context graphs
  tickernotch: Tickers, news, weather and social counters beside the notch or in the menu bar
  tight-studio: Screen recorder and video editor
  token-monitor: Monitor token usage, costs, and limits across AI coding tools
  tone3000: Amp modeling plug-in for Neural Amp Modeler captures and impulse responses
  triggerfilo: Focus timer and Kanban board for tracking tasks
  unity-cli: Command-line interface for Unity
  unity-hub@beta: Management tool for Unity
  unsloth: Desktop application for Unsloth Studio
  waku: Native desktop app for coding agents
  warp-agent-cli: Agentic development environment for command-line workflows
  work-louder-input: Keyboard configurator for Work Louder devices
  zcode: AI-assisted development environment

You have 39 outdated formulae and 1 outdated cask installed.
```

Question 2 – Git Repository and Version Control

```
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/giflib_bottle_manifest--5.2.2... (10.8KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/glib--2.86.3... (9.2MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/glib_bottle_manifest--2.86.3... (20.2KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/graphite2--1.3.14... (92.6KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/graphite2_bottle_manifest--1.3.14-2... (8.0KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/harfbuzz--12.3.2... (3.3MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/harfbuzz_bottle_manifest--12.3.2... (29.9KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/icu4c78--78.2... (31.8MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/icu4c78_bottle_manifest--78.2-1... (9.7KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/jpeg-turbo--3.1.3... (1MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/jpeg-turbo_bottle_manifest--3.1.3... (8.3KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libpng--1.6.54... (459KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libpng_bottle_manifest--1.6.54... (8.5KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libtiff--4.7.1_1... (1.9MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libtiff_bottle_manifest--4.7.1_1... (12.6KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libunistring--1.4.1... (1.9MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libunistring_bottle_manifest--1.4.1... (7.3KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libx11--1.8.12... (2.3MB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libx11_bottle_manifest--1.8.12... (18.3KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxau--1.0.12... (33.4KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxau_bottle_manifest--1.0.12... (10.3KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxcb--1.17.0... (999.8KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxcb_bottle_manifest--1.17.0-1... (19.2KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxdmcp--1.1.5... (36.8KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxdmcp_bottle_manifest--1.1.5... (12.0KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxext--1.3.7... (189.6KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxext_bottle_manifest--1.3.7... (11.5KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxrender--0.9.12... (54.6KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/libxrender_bottle_manifest--0.9.12... (14.2KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/little-cms2--2.18... (434.8KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/little-cms2_bottle_manifest--2.18... (12.2KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/loz--2.18... (149.6KB)
Removing: /Users/abhijeetsingh/Library/Caches/Homebrew/loz_bottle_manifest--2.18... (16.7KB)
```

```
4 directories, 1 file
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % pwd
/Users/abhijeetsingh/Cloud-Assignment/Abhijeet-GLA-CourseRegistration
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git init
Initialized empty Git repository in /Users/abhijeetsingh/Cloud-Assignment/Abhijeet-GLA-CourseRegistration/.git/
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  course_info.txt

nothing added to commit but untracked files present (use "git add" to track)
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git config --global user.name
Abhijeet singh
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git config --global user.email
singhabhijeet9454@gmail.com
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  course_info.txt

nothing added to commit but untracked files present (use "git add" to track)
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git config --global user.name "Abhijeet Singh"
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git config --global user.email "singhabhijeet9454@gmail.com"
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git config --global user.name
Abhijeet Singh
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git add .
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git status
On branch main

No commits yet
```

Question 3 – Feature Branch and Course Search Module

```
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   course_info.txt

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git commit -m "Initial project setup"
[main (root-commit) 14b8712] Initial project setup
 1 file changed, 4 insertions(+)
 create mode 100644 course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git log --oneline
14b8712 (HEAD -> main) Initial project setup
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % nano course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % nano course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % cat course_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
subject name:cloud computing

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % git diff
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % cat course_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
subject name:cloud computing

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % nano course_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % ^V
zsh: command not found: ^V
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-CourseRegistration % cat course_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
```

Question 4 – Branch Merge, Conflict Generation and Resolution

```
-----
+Semester: 5th
+Course/Program Name: B.Tech Computer Science
+Project Description: A course registration project for managing and searching available courses.
-----
```

```
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git add course_info.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git commit -m "Update course information"
[main 102cefc] Update course information
1 file changed, 6 insertions(+), 1 deletion(-)
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git log --oneline
102cefc (HEAD -> main) Update course information
14b8712 Initial project setup
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch course-search
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git switch course-search
Switched to branch 'course-search'
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
* course-search
main
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % cd src
abhiyeet@Abhiyeets-MacBook-Air src % pwd
/Users/abhiyeet@Abhiyeets-MacBook-Air/src
abhiyeet@Abhiyeets-MacBook-Air src % touch course_search.txt
abhiyeet@Abhiyeets-MacBook-Air src % nano course_search.txt
abhiyeet@Abhiyeets-MacBook-Air src % cat course_search.txt
Course Search Module

The Course Search Module allows students to search for available courses.
Students can search courses using course name, course code, or subject.
It helps students find relevant courses easily during course registration.
abhiyeet@Abhiyeets-MacBook-Air src % cd
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch course-search
Untracked files:
```

```
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch course-search
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/

nothing added to commit but untracked files present (use "git add" to track)
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git add src/course_search.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git commit -m "Add course search module"
[course-search 3ac9d8e] Add course search module
1 file changed, 5 insertions(+)
 create mode 100644 src/course_search.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git log --oneline
3ac9d8e (HEAD -> course-search) Add course search module
102cefc (main) Update course information
14b8712 Initial project setup
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
* course-search
main
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git switch main
Switched to branch 'main'
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
course-search
* main
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git merge course-search
Updating 102cefc..3ac9d8e
Fast-forward
 src/course_search.txt | 5 +++++
 1 file changed, 5 insertions(+)
 create mode 100644 src/course_search.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % ls src
course_search.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git log --oneline --graph --all
* 3ac9d8e (HEAD -> main, course-search) Add course search module
* 102cefc Update course information
* 14b8712 Initial project setup
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git switch -c course-update
Switched to a new branch 'course-update'
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
course-search
* course-update
main
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % nano course_info.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git add course_info.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git commit -m "Update project name in course-update"
[course-update 4a2e0ec] Update project name in course-update
1 file changed, 1 insertion(+), 1 deletion(-)
-----
+Semester: 5th
+Course/Program Name: B.Tech Computer Science
+Project Description: A course registration project for managing and searching available courses.
-----
```

```
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % nano course_info.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % cat course_info.txt
Student Name: Abhiyeet Singh
Roll Number: 2415861871
Subject Name: Cloud Computing
<<<<<< HEAD
Project Name: Advanced Course Registration system
=====
Project Name: Course Registration System
>>>>>> course-update
Semester: 5th
Course/Program Name: B.Tech Computer Science
Project Description: A course registration project for managing and searching available courses.
```

```
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git add course_info.txt
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git commit -m "Resolve merge conflict"
[main a48f7bb] Resolve merge conflict
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git log --oneline --graph --all
|
| * a48f7bb (HEAD -> main) Resolve merge conflict
|/
| * 4a2e0ec (course-update) Update project name in course-update
| * 190bdf9 Update project name in main
|/
* 3ac9d8e (course-search) Add course search module
* 102cefc Update course information
* 14b8712 Initial project setup
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
course-search
course-update
* main
abhiyeet@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration %
```

Question 5 – Final Demonstration

Project Description: A course registration project for managing and searching available courses.

```
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git add course_info.txt
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git commit -m "Resolve merge conflict"
[main a48f7bb] Resolve merge conflict
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git log --oneline --graph --all
* a48f7bb (HEAD -> main) Resolve merge conflict
|
| * 4a2eeec (course-update) Update project name in course-update
| | 198bdf9 Update project name in main
|/
* 3ac9d8e (course-search) Add course search module
* 102cefc Update course information
* 14b8712 Initial project setup
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % git branch
course-search
course-update
* main
abhiyeetsingh@Abhiyeets-MacBook-Air Abhiyeet-GLA-CourseRegistration % █
```